WEEK 5 – POWER BI HEALTHCARE PATIENT DATA ANALYSIS

Name: Monisha J

Roll No: E25DA029

Department: B.Sc DS & AI

INTRODUCTION:

This week, healthcare patient data was prepared and organized using Power BI and Power Query. The original cleaned dataset was used to create separate tables for patient details, admission details, and billing information. Columns were selected according to their purpose, and a new Age Group column was created for patient analysis.

DATA CLEANING AND PREPARATION:

The healthcare dataset was cleaned before creating the required tables. Duplicate records were removed to keep the data accurate. The Name column was also standardized to maintain consistency. After cleaning the data, the cleaned query was used to create the Dim_Patient, Dim_Admission, and Billing tables.

Name	Age	Gender
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
Bobby Jackson	30	Male
Leslie Terry	62	Male
Danny Smith	76	Female
Andrew Watts	28	Female
Adrienne Bell	43	Female
Emily Johnson	36	Male
Edward Edwards	21	Female
Christina Martinez	20	Female
Jasmine Aguilar	82	Male
Christopher Berg	58	Female
Michelle Daniels	72	Male

CREATING PATIENT_ID AND ADMISSION_ID:

An Index Column was added to the original cleaned query using Add Column → Index Column. The Index column was renamed to Patient ID. This created a unique identifier for each patient record.

Patient _ID	Admission_ID
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11

CREATING DIM_PATIENT TABLE

A Reference of the original cleaned query was created and renamed as Dim_Patient. Using Choose Columns, only the required patient-related fields were retained:

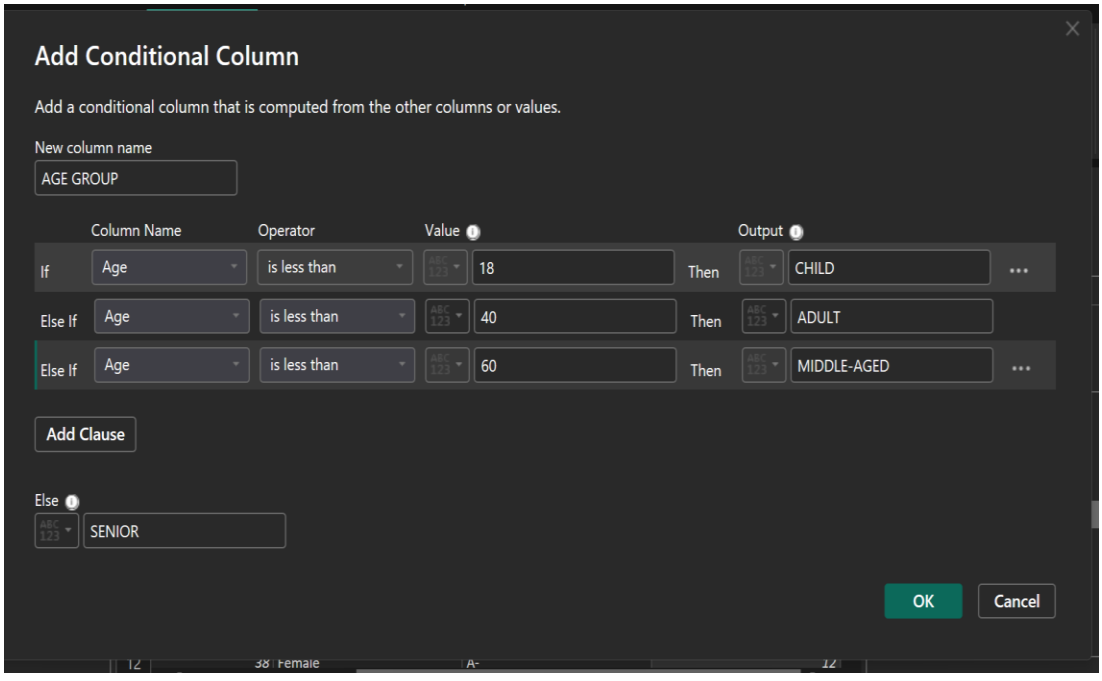
- Patient ID
- Name
- Age
- Blood Type
- Gender

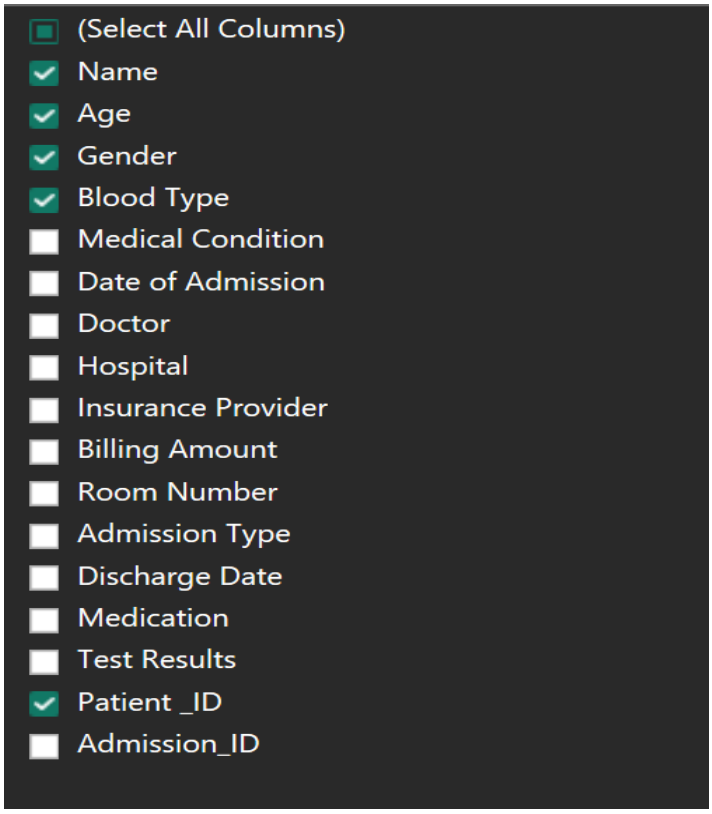
A conditional column named Age Group was then created.

The Age Group rules were:

- Age < 18 → Child
- Age < 40 → Adult
- Age < 60 → Middle-Aged
- Otherwise → Senior

Therefore, the final Dim_Patient table contains six columns.





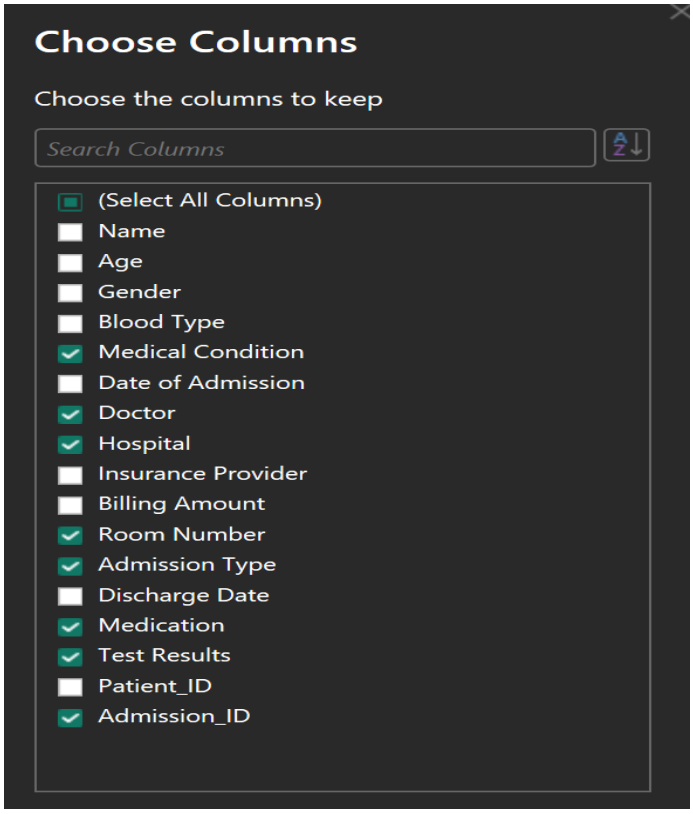
CREATING DIM ADMISSION TABLE:

A new Reference was created from the original cleaned query and renamed as Dim_Admission.

The following eight columns were retained:

- Admission ID
- Admission Type
- Doctor
- Hospital
- Medical Condition
- Medication
- Room Number
- Test Results

Unwanted columns were removed using Choose Columns.

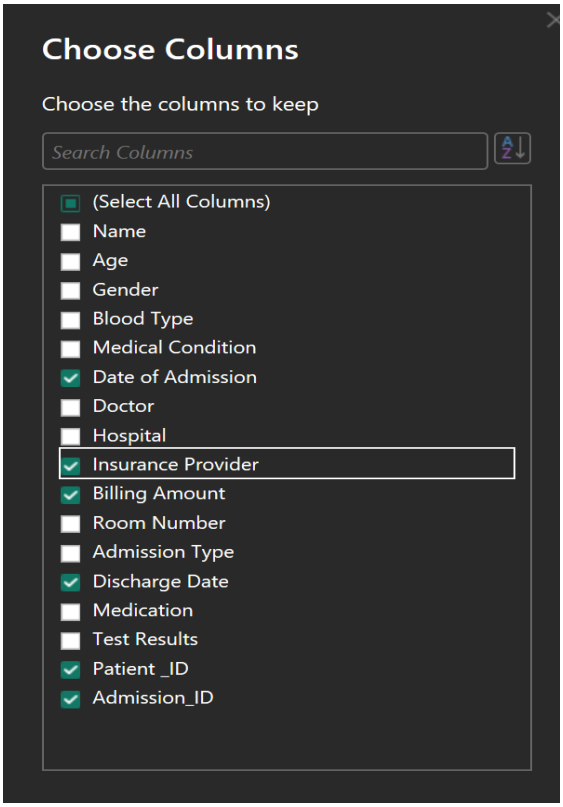


CREATING BILLING TABLE:

Another Reference was created from the original cleaned query and renamed as Billing.

The following six columns were retained:

- Admission ID
- Patient ID
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider



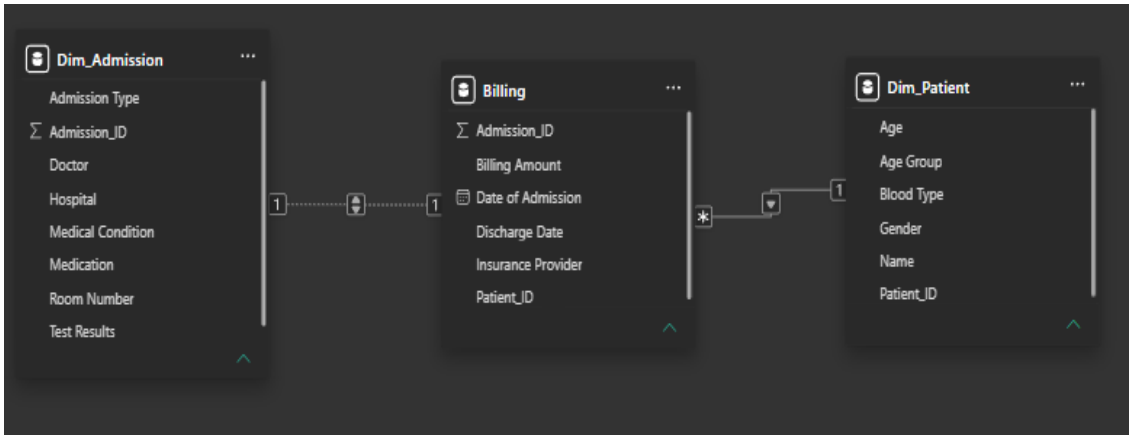
DATA MODEL:

After creating the three tables, Close & Apply was selected to load them into the Power BI model.

The final model contains:

- Dim_Patient – patient information and age groups
- Dim_Admission – admission and medical information
- Billing – billing and date information

  The tables were connected using Patient ID and Admission ID.



CONCLUSION:

The Week 5 healthcare data was prepared using Power Query and Power BI. Three tables were created: Dim_Patient, Dim_Admission, and Billing. Patient ID and Age Group were also created. Unwanted columns were removed, and the final tables were added to the Power BI model.